

THEME & DESIGN SYSTEM — MONOCHROME UI SPECIFICATION

The StockSense design system

A strict black-and-white visual language for the StockSense Streamlit dashboard: typography, spacing, colour, components, charts, and layout — specified precisely enough to build from.

PALETTE

Grayscale only — no
accent colour

TYPEFACES

Space Grotesk · Inter ·
JetBrains Mono

REFERENCE

Minimal, product-grade AI UI
(ChatGPT-style)

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01 Design

Philosophy

StockSense outputs a probability-weighted judgement about somebody's money. The interface's only job is to make that judgement legible, calm, and honest — never persuasive, never noisy.

1.1 Why black and white, deliberately

Colour is the easiest way to fake confidence in a dashboard — a green UP badge feels more certain than a black one, even when the underlying probability is identical. StockSense strips colour out entirely and works in a pure grayscale system so that the only thing carrying meaning is **typography, weight, spacing, and position** — never hue. This keeps the SHAP explanation, not the paint job, as the thing that earns the user's trust.

1.2 Reference point: minimal AI-product UI

The visual reference for this system is the current generation of minimal, text-forward AI product interfaces — most visibly ChatGPT's own product design: near-white and near-black surfaces, restrained borders instead of drop shadows, generous line-height, one accent-free type system doing all the work, and interface chrome that recedes so the content (here: price data, sentiment, predictions, SHAP values) stays the loudest thing on screen.

1.3 Four principles

01 – LEGIBILITY OVER DECORATION

Every visual choice must make a number, a label, or a trend easier to read — or it doesn't ship.

02 – GRAYSCALE-ONLY MEANING

No colour-coded semantics. Direction, confidence, and risk are shown with weight, fill, and position instead.

03 – CALM DENSITY

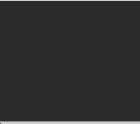
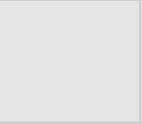
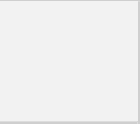
Fit real analytical density (SHAP tables, indicator series) without ever feeling cluttered — whitespace is a component, not a leftover.

04 – EXPLAINABILITY IS A UI CITIZEN

The SHAP panel is designed with the same weight as the prediction itself, never demoted to a footnote or tooltip.

02 Colour System

One achromatic ramp, nine stops. No blue, no green, no red — direction is communicated through shape and label, not hue.

								
Ink #0A0A0A	Ink / 2 #2B2B2B	Gray / 1 #4A4A4A	Gray / 2 #767676	Gray / 3 #A3A3A3	Gray / 4 #CFCFCF	Gray / 5 #E6E6E6	Gray / 6 #F2F2F2	Paper #FFFFFF

2.1 Usage rules

TOKEN	USED FOR	NEVER USED FOR
Ink #0A0A0A	Primary text, headings, primary buttons, active states	Large background fills
Gray/1-2	Secondary text, chart lines, icons	Body copy under 8pt
Gray/3-4	Borders, dividers, disabled states, gridlines	Any text meant to be read closely
Gray/5-6	Card backgrounds, hover fills, table zebra	Borders (too low-contrast)
Paper #FFFFFF	Base canvas	—

2.2 How direction is shown without colour

  	Fill vs. outline vs. muted — not colour — carries the meaning.
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03 Typography System

Three typefaces, each with exactly one job — this document is set in the same system it specifies.

TYPEFACE	ROLE	WHY
Space Grotesk	Display / headings	A geometric grotesk with just enough character to feel like a product, not a spreadsheet — used for H1–H4, KPI numbers, and card titles only.
Inter	Body / UI text	Neutral, highly legible at small sizes, huge weight range — used for all paragraph copy, labels, and dashboard microcopy.
JetBrains Mono	Data / code	Tabular figures and a coding-tool feel — used for tickers, feature names, SHAP values, timestamps, and every raw number in a table.

3.1 Type scale

Aa Stock Direction

Display / H1
Space Grotesk 700
26pt / 1.1 / -0.01em

Aa Prediction Panel

H2 – Section title
Space Grotesk 700
18pt / 1.15

Aa SHAP Contribution

H3 – Card title
Space Grotesk 600
13pt / 1.2

Aa The model's next-day call is UP, driven primarily by sentiment.

Body
Inter 400
10pt / 1.55

Aa CONFIDENCE • UPDATED 09:31

Caption / Label
Inter 500, uppercase
8.5pt / 0.06em tracking

RSI_14 = 71.4 Δ +0.083

Data / Mono

JetBrains Mono 400-500

10pt / tabular figures

3.2 Weight ramp (Inter)

300 Light — reserved for large display numbers only

400 Regular — default body copy

500 Medium — UI labels, table headers

600 Semibold — emphasis inside body copy, card titles

700 Bold — headings, KPI figures only

Rule of thumb: if you reach for bold outside a heading or KPI, use semibold instead — bold is reserved so it keeps its weight, literally.

04

Spacing & Layout Grid

An 8pt base unit keeps every card, gutter, and touch target on the same rhythm.

4.1 Spacing scale

space-1		4pt
space-2		8pt
space-3		16pt
space-4		24pt
space-5		32pt
space-6		48pt
space-7		64pt

4.2 Dashboard grid

The Streamlit dashboard runs on a **12-column grid** inside a fixed sidebar + fluid main-area shell: a 220px sidebar for ticker selection, date range, and model settings; a fluid main canvas using space-4 (24pt) gutters between cards; card corner radius of **8px**; border weight of **1px** at Gray/4.

REGION	WIDTH	NOTES
Sidebar	220px fixed	Ticker, date range, model/version selector, refresh control.
Main canvas	fluid, max 1200px	12-col grid, 24pt gutters, 32pt outer margin.
Card radius	8px	Applied to all panels, badges use pill radius (999px).
Card border	1px, Gray/4	No drop shadows — flat, bordered surfaces only.

05 Iconography & Imagery

Icons are structural, not decorative — every glyph on the dashboard has a specific analytical job.

5.1 Style rules

- **Stroke-only, 1.5px weight** — no filled icon glyphs except direction indicators (▲ / ▼), which use fill deliberately for at-a-glance scanning.
- **16px / 20px sizes only** — 16px inline with text, 20px standalone in card headers.
- **Ink or Gray/2** — never a third tone; icons match the text they sit beside.
- **No skeuomorphism, no gradients, no shadows** on any glyph.

5.2 Chart glyph set

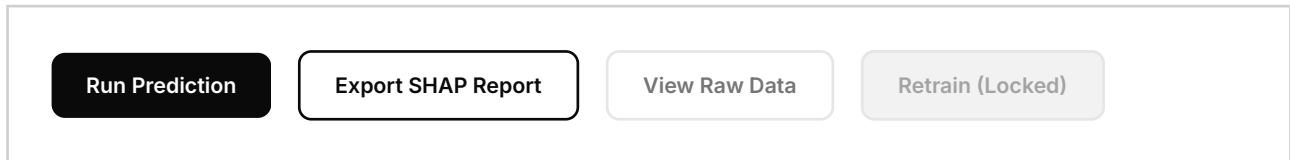
Three geometric primitives are used consistently across every chart in the dashboard so a returning user reads them instantly:

- ▲ Directional gain / UP prediction
- ▼ Directional loss / DOWN prediction
- Point-in-time marker on a trend line (news event, prediction day)

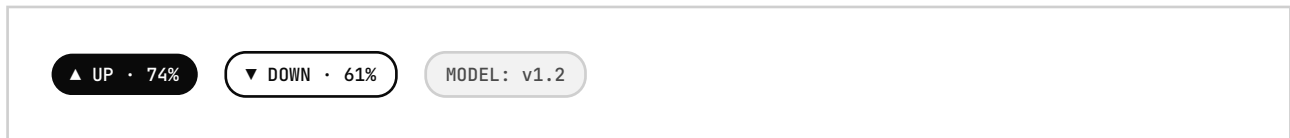
06 Component Library

The full set of interface primitives the Streamlit dashboard is assembled from.

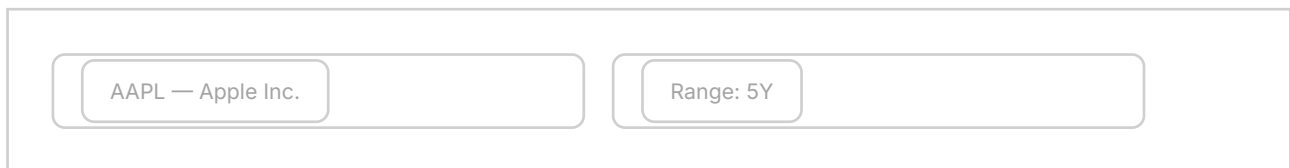
6.1 Buttons



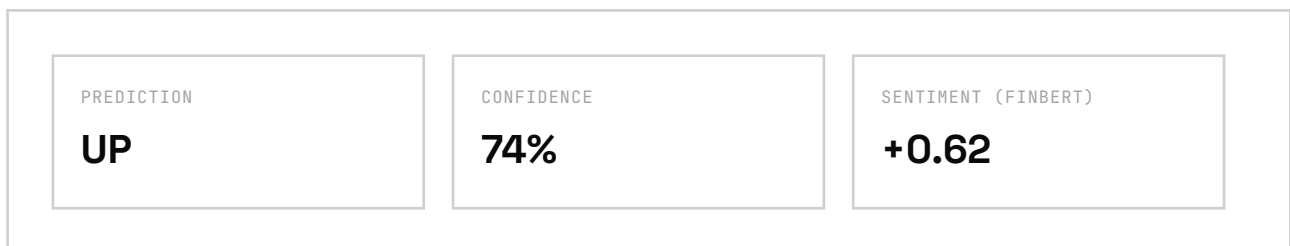
6.2 Status badges



6.3 Inputs & selectors



6.4 KPI / stat cards



6.5 Tables

Dashboard tables (SHAP rankings, indicator readouts) reuse this document's own table style exactly: mono uppercase headers, hairline row dividers, no vertical rules, numeric columns right-aligned and set in JetBrains Mono.

07

Data Visualisation

Style

Every chart on the dashboard — price trend, sentiment series, SHAP bars — follows one grayscale charting grammar.

CHART ELEMENT	STYLE RULE
Primary series (price / prediction)	Solid line, 2px, Ink #0A0A0A.
Secondary series (moving average)	Dashed line, 1.5px, Gray/2.
Sentiment series	Stepped/area line at 12% fill opacity of Ink, so it recedes behind price.
SHAP bars	Horizontal bars: filled Ink bar = positive (pushes UP), outlined bar = negative (pushes DOWN); length encodes magnitude.
Gridlines	Gray/5, 0.5px, horizontal only — no vertical gridlines.
Axis labels	JetBrains Mono, 9px, Gray/2.
Tooltips	Ink background, Paper text, 6px radius, no shadow — a solid "card" that appears on hover.

7.1 SHAP bar chart — reference render

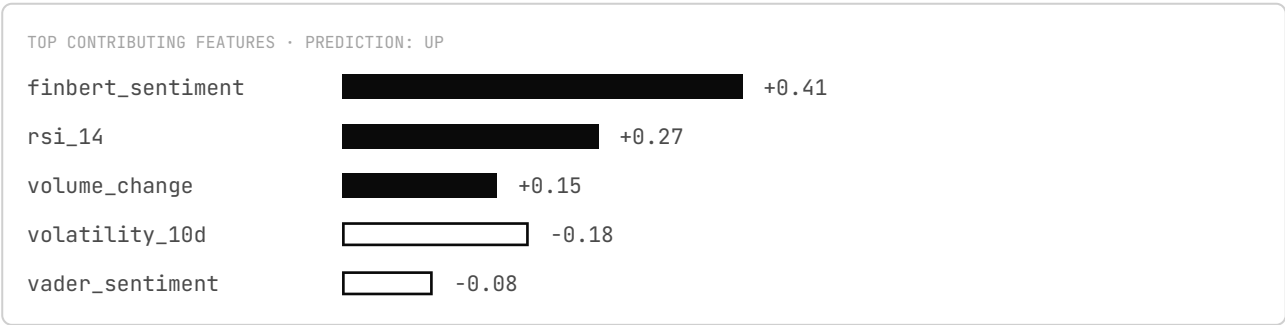


FIG. 7.1 — FILLED BARS PUSH THE PREDICTION TOWARD UP; OUTLINED BARS PULL IT TOWARD DOWN. BAR LENGTH IS PROPORTIONAL TO |SHAP VALUE|.

08 Dashboard Layout Blueprint

The complete Streamlit screen, wireframed to scale, showing exactly where the four panels from the project report (Trends · Sentiment · Prediction · SHAP) sit.



FIG. 8.1 – STOCKSENSE DASHBOARD BLUEPRINT: SIDEBAR CONTROLS, PRICE/INDICATOR TREND, PREDICTION CARD, SENTIMENT SERIES, AND SHAP EXPLANATION, ALL ON ONE SCREEN.

09 Interaction & Motion

Motion is used to orient, never to entertain — this is a decision-support tool, not a marketing surface.

STATE	BEHAVIOUR
Hover (card / row)	Background steps to Gray/6; no scale, no shadow.
Active (button)	Ink darkens by one step; 80ms linear transition.
Loading (model run)	A single 1px indeterminate bar in Ink along the card's top edge — no spinners, no skeleton shimmer.
Chart tooltip	Appears on hover only, fades in over 100ms, no delay on dismiss.
Disabled	Gray/6 fill, Gray/3 text, no hover response at all.

10 Accessibility & Contrast

- Body text (Ink on Paper) sits at a 19.5:1 contrast ratio — comfortably above WCAG AAA (7:1) for normal text.
- The lightest permitted text tone, Gray/2 on Paper, is reserved for captions $\geq$ 8pt and still clears AA (4.5:1).
- Direction is never conveyed by colour alone — badges, icons (▲/▼), and text labels always co-occur, so the system remains legible for colour-vision-deficient users even though it is already grayscale.
- Minimum interactive target size on the dashboard: 32×32px, spaced at least 8pt apart.

11 Do's and Don'ts

DO

- Use weight and fill to encode meaning
- Keep every chart on the grayscale grammar (Sec. 07)
- Give the SHAP panel equal visual weight to the prediction
- Use hairline borders instead of shadows

DON'T

- Introduce green/red for UP/DOWN
- Use more than three font weights on one screen
- Add drop shadows, gradients, or skeuomorphic icons
- Let decorative charts outrank the SHAP explanation

12 Design Token Reference

The complete token set, ready to drop into a Streamlit theme config or a CSS variables file.

TOKEN	VALUE	NOTES
color.ink	#0A0A0A	Primary text & primary surfaces
color.gray.1	#4A4A4A	Secondary text
color.gray.2	#767676	Tertiary text, chart axis labels
color.gray.3	#A3A3A3	Disabled text, secondary chart lines
color.gray.4	#CFCFCF	Borders, dividers
color.gray.5	#E6E6E6	Gridlines
color.gray.6	#F2F2F2	Card / hover backgrounds
color.paper	FFFFFF	Base canvas
font.display	Space Grotesk	H1–H4, KPI figures
font.body	Inter	Paragraphs, labels, UI text
font.mono	JetBrains Mono	Tickers, SHAP values, timestamps
radius.card	8px	All panels and cards
radius.pill	999px	Badges, tags
space.1 – space.7	4 / 8 / 16 / 24 / 32 / 48 / 64px	Full spacing scale, Section 04
border.default	1px solid gray. 4	Standard card / table border

This document is itself typeset entirely in the tokens defined above — it is both the specification and a working proof of the system.